

Willmore inequalities for hypersurfaces with boundary

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(joint works with Xiaohan Jia, Guofang Wang, Chao Xia)

- 1 Willmore inequalities: an overview of the history
- 2 Willmore inequality for embedded hypersurfaces in arbitrary convex domain
- 3 Willmore inequality for capillary immersions of surfaces

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Willmore inequality for closed surfaces

Willmore energy

- Given a closed (compact, without boundary) surface Σ immersed in $\mathbf{R}^3$ through the map F , Willmore associated to every such surface a quantity nowadays known as the *Willmore energy*

$$\mathcal{W}(F) = \frac{1}{4} \int_{\Sigma} H^2 d\mu_g = \frac{1}{4} \int_{\Sigma} (\kappa_1 + \kappa_2)^2 d\mu_g,$$

where κ_1, κ_2 are the principal curvatures of Σ , $H = (\kappa_1 + \kappa_2)$ is the mean curvature, and $d\mu_g$ is the area element of Σ induced by the immersion F .

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- This energy, which Willmore was not initially aware of, had appeared in the works of Blaschke and his student Thomsen back in the 1920s.

Willmore inequality for closed surfaces

Willmore inequality

- Willmore showed that round spheres have the least possible Willmore energy among all immersions of closed surfaces in $\mathbf{R}^3$, which follows from the establishment of the well-known Willmore inequality: for any immersion F of closed surface Σ in $\mathbf{R}^3$,

$$\mathcal{W}(F) \geq 4\pi,$$

with equality holds if and only if $F(\Sigma)$ is a round sphere.

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with equality holds if and only if $F(\Sigma)$ is a round sphere.

- Willmore himself then tried to find the minimizing shape among the class of tori (genus 1 closed surfaces) and proposed the famous *Willmore conjecture*:

Every compact surface Σ of genus 1 in $\mathbf{R}^3$ satisfies

$$\mathcal{W}(F) \geq 2\pi^2,$$

which was answered by Marques-Neves¹ using Min-Max method.

¹Fernando C. Marques and André Neves, *Min-max theory and the Willmore conjecture*, *Ann. of Math.*(2) 179 (2014)

Willmore inequality for closed surfaces

Conformal invariance

We recall several classical ways to prove the Willmore inequality.

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Conformal point of view.

- Willmore energy for closed surfaces is *conformal invariant* (in $\mathbf{R}^{n+1}$ for $n \geq 2$) since it is invariant under *rigid motions*, *scalings*, and *spherical inversion* $x \mapsto \frac{x}{|x|^2}$.
- The conformal invariance can be also deduced by using Gauss-Bonnet:

$$\begin{aligned}\mathcal{W}(F) &= \frac{1}{4} \int_{\Sigma} (\kappa_1 + \kappa_2)^2 d\mu_g = \frac{1}{2} \int_{\Sigma} \frac{(\kappa_1 - \kappa_2)^2}{2} d\mu_g + \int_{\Sigma} \kappa_1 \kappa_2 d\mu_g \\ &= \frac{1}{2} \int_{\Sigma} |\mathring{A}|^2 d\mu_g + 2\pi\chi(\Sigma),\end{aligned}$$

where $\chi(\Sigma)$ is the Euler characteristic of Σ which is topological invariant and the quantity $|\mathring{A}|^2 d\mu_g$ is conformal invariant since Σ is 2-dimensional.

Li-Yau inequality

- Li and Yau ¹ were the first to exploit in a crucial way the conformal invariance of Willmore energy and they proved the nowadays so-called *Li-Yau inequality*:

$$\mathcal{W}(F) \geq 4\pi\Theta^2(\mu_g, p)$$

for any $p \in F(\Sigma)$. Here $\Theta^2(\mu_g, p)$ is the *density* of the immersion at p , namely, assume $\Theta^2(\mu_g, p) = k \in \mathbb{N}$ then nearby p the surface behaves like k -small disks containing p .

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- Clearly $k \geq 1$ and hence we obtain $\mathcal{W}(F) \geq 4\pi$ as desired. Moreover, as a by-product, it also follows that if the Willmore energy of any immersion of closed surfaces is less than 8π , then it must be an embedding.

¹Peter Li and Shing-Tung Yau, *A new conformal invariant and its applications to the Willmore conjecture and the first eigenvalue of compact surfaces*, *Invent. Math.* 69 (1982).

Simon's Monotonicity identity

Simon's Monotonicity identity.

- For a closed surface Σ embedded/immersed in $\mathbf{R}^{n+1}$ for $n \geq 2$, it is proven by Simon ¹ / Kuwert-Schätzle ² that for $0 < \sigma < \rho < \infty$ and $a \in \mathbf{R}^{n+1}$,

$$g_a(\rho) - g_a(\sigma) = \frac{1}{\pi} \int_{\Sigma \cap B_\rho(a) \setminus B_\sigma(a)} \left| \frac{1}{4} \vec{\mathbf{H}} + \frac{(x-a)^\perp}{|x-a|^2} \right|^2 d\mathcal{H}^2,$$

where

$$\begin{aligned} g_a(r) := & \frac{\mathcal{H}^2(\Sigma \cap B_r(a))}{\pi r^2} + \frac{1}{16\pi} \int_{\Sigma \cap B_r(a)} |\vec{\mathbf{H}}|^2 d\mathcal{H}^2 \\ & + \frac{1}{2\pi r^2} \int_{\Sigma \cap B_r(a)} \vec{\mathbf{H}} \cdot (x-a) d\mathcal{H}^2. \end{aligned}$$

¹Leon Simon, *Existence of surfaces minimizing the Willmore functional*, *Comm. Anal. Geom.*1 (1993)

²Ernst Kuwert and Reiner Schätzle, *Removability of point singularities of Willmore surfaces*, *Ann. of Math.*(2) 160 (2004)

Simon's Monotonicity identity

An alternative proof of Willmore inequality and Li-Yau inequality

- Simon's monotonicity identity yields an alternative approach to Willmore inequality and Li-Yau inequality, since we could let $\sigma \searrow 0$ and $\rho \rightarrow \infty$ to obtain

$$\frac{1}{4\pi} \mathcal{W}(F) = g_a(\infty) \geq g_a(0) \stackrel{\text{Hölder ineq}}{=} \Theta^2(\mu_g, a),$$

where we recall

$$\begin{aligned} g_a(r) := & \frac{\mathcal{H}^2(\Sigma \cap B_r(a))}{\pi r^2} + \frac{1}{16\pi} \int_{\Sigma \cap B_r(a)} |\vec{\mathbf{H}}|^2 d\mathcal{H}^2 \\ & + \frac{1}{2\pi r^2} \int_{\Sigma \cap B_r(a)} \vec{\mathbf{H}} \cdot (x - a) d\mathcal{H}^2. \end{aligned}$$

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Rigidity characterization

- The equality case can be easily characterized: **1.** the immersion is in fact an embedding.

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$$\int_{\Sigma} \left| \frac{1}{4} \vec{H} + \frac{(x-a)^{\perp}}{|x-a|^2} \right|^2 d\mathcal{H}^2 = 0, \quad \forall a \in \Sigma.$$

Fix any $a \in \Sigma$, by continuity the above equality implies

$$\frac{1}{4} \vec{H}(x) + \frac{(x-a)^{\perp}}{|x-a|^2} = 0, \quad \forall x \in \Sigma.$$

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Fix any $a \in \Sigma$, by continuity the above equality implies

$$\frac{1}{4} \vec{\mathbf{H}}(x) + \frac{(x-a)^{\perp}}{|x-a|^2} = 0, \quad \forall x \in \Sigma.$$

- 3.** Since Σ is closed, embedded in $\mathbf{R}^{n+1}$, there exists some elliptic point $x_0 \in \Sigma$. Let $x = x_0$ in the above equality, after rearranging one finds

$$\left| a - \left(x_0 - \frac{2\nu(x_0)}{|\vec{\mathbf{H}}(x_0)|} \right) \right|^2 = \frac{4}{|\vec{\mathbf{H}}(x_0)|^2}, \quad \forall a \in \Sigma,$$

which implies spherical.

Simon's Monotonicity identity

Sketch of proof: varifolds formulation

- Let $F : \Sigma \rightarrow \mathbf{R}^{n+1}$ be an immersion of closed surface. Write $g = F^*g_{\text{euc}}$ the pull-back metric and μ_g the induced area measure. Its image as (integral) varifold is given by

$$\mu := F(\mu_g) = \theta \mathcal{H}^2 \llcorner F(\Sigma), \quad \theta(x) = \mathcal{H}^0(F^{-1}\{x\}).$$

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- The first variation formula of μ : for any $\phi \in C_c^1(\mathbf{R}^{n+1}; \mathbf{R}^{n+1})$

$$\int \operatorname{div}_\mu \phi \, d\mu = - \int \langle \vec{\mathbf{H}}, \phi \rangle \, d\mu,$$

where $\vec{\mathbf{H}}$ is the generalized mean curvature of μ which is square integrable; $\operatorname{div}_\mu \phi$ denotes the tangential divergence with respect to the approximate tangent space of μ .

Simon's Monotonicity identity

Sketch of proof

- Fix any $a \in \mathbf{R}^{n+1}$, $0 < \sigma < \rho < \infty$, the Simon's Monotonicity identity follows after testing the first variation of μ by the vector field

$$\phi(x) := \left(\frac{1}{|x - a|_\sigma^2} - \frac{1}{\rho^2} \right)_+ (x - a),$$

where $|x - a|_\sigma = \max\{|x - a|, \sigma\}$.

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where $|x - a|_\sigma = \max\{|x - a|, \sigma\}$.

- ϕ is Lipschitz.
- In $B_\sigma(a)$, $\phi(x) = \left(\frac{1}{\sigma^2} - \frac{1}{\rho^2} \right)(x - a)$;
In $B_\rho(a) \setminus B_\sigma(a)$, $\phi(x) = \left(\frac{1}{|x - a|_\sigma^2} - \frac{1}{\rho^2} \right)(x - a)$;
Outside $B_\rho(a)$, ϕ vanishes.

Simon's Monotonicity identity

Sketch of proof

- Recall

$$g_a(r) := \frac{\mathcal{H}^2(\Sigma \cap B_r(a))}{\pi r^2} + \frac{1}{16\pi} \int_{\Sigma \cap B_r(a)} |\vec{\mathbf{H}}|^2 d\mathcal{H}^2 \\ + \frac{1}{2\pi r^2} \int_{\Sigma \cap B_r(a)} \vec{\mathbf{H}} \cdot (x - a) d\mathcal{H}^2,$$

to obtain this we observe that on $B_\rho(a) \setminus B_\sigma(a)$,

$$\operatorname{div}_\mu \phi(x) = -\frac{2}{\rho^2} + 2 \left| \frac{(x-a)^\perp}{|x-a|^2} \right|^2.$$

While we significantly use the fact that $\vec{\mathbf{H}} \perp F(\Sigma)$ for μ -a.e., so that for any vector $v \in \mathbf{R}^{n+1}$, one has (in fact, we choose $v = \frac{(x-a)^\perp}{|x-a|^2}$)

$$2 \left| \frac{1}{4} \vec{\mathbf{H}} + v^\perp \right|^2 = \frac{1}{8} |\vec{\mathbf{H}}|^2 + 2|v^\perp|^2 + \vec{\mathbf{H}} \cdot v.$$

Simon's Monotonicity identity

Generalization to the context of varifolds

- To generalize the above argument to work for general varifolds, we just have to make sure that the generalized mean curvature vector $\vec{H}$ is perpendicular to the approximate tangent space, which is ensured by Brakke's perpendicular theorem ¹, provided that μ is integral.

¹Kenneth A. Brakke, *The motion of a surface by its mean curvature*, PhD Thesis, Princeton University, (1978)

²Alexander Volkmann, *A monotonicity formula for free boundary surfaces with respect to the unit ball*, *Comm. Anal. Geom.* 24 (2016)

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- A similar result is proven for free boundary immersions of surfaces with boundary/free boundary integral 2-varifolds in Euclidean unit ball by Volkmann ².

¹Kenneth A. Brakke, *The motion of a surface by its mean curvature*, PhD Thesis, Princeton University, (1978)

²Alexander Volkmann, *A monotonicity formula for free boundary surfaces with respect to the unit ball*, *Comm. Anal. Geom.* 24 (2016)

Willmore inequality for submanifolds of any co-dimensions

- Chen ¹ proved the following Willmore inequality for n -dimensional closed submanifold Σ immersed in $\mathbf{R}^{n+m}$ ($m \geq 1$):

$$\int_{\Sigma} |H|^n d\mu_g \geq c = c(n),$$

with equality holds if and only if Σ is embedded as a hypersphere in an $(n+1)$ -dimensional linear subspace of $\mathbf{R}^{n+m}$ if $n \geq 2$.

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- The integral on the left is the Willmore energy of submanifolds. Note that it is not conformal invariant unless $n = 2$.

¹Bang-Yen Chen, *On a theorem of Fenchel-Borsuk-Willmore-Chern-Lashof*, *Math. Ann.* 194 (1971)

Willmore inequality for submanifolds of any co-dimensions

- Later Heintze-Karcher ¹ gave an alternative proof of Chen's result. More precisely, they obtained: For a compact submanifold Σ^n in a Riemannian manifold M^{n+m} . If the curvature of M is bounded from below by some positive constant $\delta > 0$, then there holds

$$\text{Vol}(M^{n+m}) \leq \frac{\text{Vol}(S_\delta^{n+m})}{\text{Vol}(S_1^n)} \int_\Sigma (\delta + H^2)^{\frac{n}{2}} d\mu_g,$$

where S_δ^{m+n} denotes sphere of dimension $m+n$ with curvature δ .

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- Willmore inequality then follows from a limit argument $\delta \searrow 0$.

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- Willmore inequality then follows from a limit argument $\delta \searrow 0$.
- However, using this original argument one cannot obtain rigidity of the inequality. Imprecisely speaking, this is because of the lack of characterization of umbilical submanifolds (*Nabelpunktsatz*) in higher co-dimensions.

¹Ernst Heintze and Hermann Karcher, *A general comparison theorem with applications to volume estimates for submanifolds*, *Ann. Sci. École Norm. Sup.* (4)11 (1978) 

Willmore inequality for embedded hypersurfaces

Geodesic normal flow: sketch of proof

Heintze-Karcher's comparison argument is powerful in the co-dimension 1 case.

- (settings) For any embedded closed hypersurface $\Sigma \subset \mathbf{R}^{n+1}$ which encloses a bounded domain Ω . The geodesic normal flow is defined as

$$\zeta(x, t) = x + t\nu(x), \quad (x, t) \in \Sigma \times \mathbf{R}_+ =: Z.$$

- (distance function) Consider the distance function to $\overline{\Omega}$ defined as

$$d(y, \overline{\Omega}) := \inf_{r>0} \left\{ r : \overline{B_r(y)} \cap \overline{\Omega} \neq \emptyset \right\}.$$

Willmore inequality for embedded hypersurfaces

Geodesic normal flow: sketch of proof

- **(Step 1)** For any $R > 0$, we have the crucial inclusion:

$$\begin{aligned} & \{y \in \mathbf{R}^{n+1} \setminus \overline{\Omega} : d(y, \overline{\Omega}) \leq R\} \\ & \subset \{\zeta(x, t) \mid x \in \Sigma, t \in (0, \min(R, \tau(x)))\}, \end{aligned}$$

where τ is a function defined on Σ by

$$\tau(x) = \begin{cases} +\infty, & \text{if } \kappa_i(x) \geq 0 \text{ for any } i = 1, \dots, n, \\ -\frac{1}{\min_i \kappa_i(x)}, & \text{otherwise.} \end{cases}$$

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This follows simply by using the foliation of concentric balls centered at y .

Willmore inequality for embedded hypersurfaces

Geodesic normal flow: sketch of proof

- **(Step 2)** Asymptotic volume ratio:

$$\lim_{R \rightarrow +\infty} \frac{\text{Vol}(\{y \in \mathbf{R}^{n+1} : d(y, \overline{\Omega}) \leq R\})}{|B_1| R^{n+1}} = 1.$$

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This can be seen easily since $\overline{\Omega}$ is bounded, for R large enough it behaves like a point.

Willmore inequality for embedded hypersurfaces

Geodesic normal flow: sketch of proof

- **(Step 3)** Estimate the volume using area formula:

$$\begin{aligned} & \text{Vol}(\{y \in \mathbf{R}^{n+1} : d(y, \overline{\Omega}) \leq R\}) \\ & \leq |\Omega| + \int_{\Sigma} \int_0^{\min(R, \tau(x))} J^Z \zeta(x, t) dt d\mu_g, \end{aligned}$$

where

$$J^Z \zeta(x, t) = \prod_{i=1}^n (1 + t \kappa_i(x)) \stackrel{\text{AM-GM}}{\leq} \left(1 + \frac{H(x)}{n} t\right)^n.$$

Willmore inequality for embedded hypersurfaces

Geodesic normal flow: sketch of proof

- Divide Σ into two parts $\Sigma_+ := \{x \in \Sigma : H(x) > 0\}$ and $\Sigma \setminus \Sigma_+$. On $\Sigma \setminus \Sigma_+$, we have

$$0 \leq \left(1 + \frac{H(x)}{n}t\right)^n \leq 1, \quad \forall t \in [0, \tau(x)],$$

which gives

$$\int_{\Sigma \setminus \Sigma_+} \int_0^{\min(R, \tau(x))} J^Z \zeta_F(x, t) dt dA \leq O(R).$$

Willmore inequality for embedded hypersurfaces

Geodesic normal flow: sketch of proof

- **(Step 4)** Combining the listed facts, we obtain:

$$\begin{aligned} & \text{Vol}(\{y \in \mathbf{R}^{n+1} : d(y, \overline{\Omega}) \leq R\}) \\ & \leq |\Omega| + \int_{\Sigma_+} \int_0^{\min(R, \tau(x))} J^Z \zeta \, dt \, d\mu_g + \int_{\Sigma \setminus \Sigma_+} \int_0^{\min(R, \tau(x))} J^Z \zeta \, dt \, d\mu_g \\ & \leq \int_{\Sigma_+} \int_0^R \left(1 + \frac{H(x)}{n} t\right)^n \, dt \, d\mu_g + O(R) \\ & = \frac{R^{n+1}}{n+1} \int_{\Sigma_+} \left(\frac{H(x)}{n}\right)^n \, d\mu_g + O(R^n) \leq \frac{R^{n+1}}{n+1} \int_{\Sigma} \left|\frac{H}{n}\right|^n \, d\mu_g + O(R^n). \end{aligned}$$

Divided both sides by R^{n+1} , then letting $R \rightarrow \infty$ and invoking the AVR, we obtain

$$|S_1^n| = (n+1)|B_1^{n+1}| \leq \int_{\Sigma} \left|\frac{H}{n}\right|^n \, d\mu_g.$$

Willmore inequality for embedded hypersurfaces

Geodesic normal flow: sketch of proof

- **(Step 5) Rigidity:**

First we could show that $\text{Vol}(\Sigma \setminus \Sigma_+) = 0$ otherwise the equality case is never achieved. This implies, by continuity, $\Sigma = \Sigma_+$ is mean convex.

Willmore inequality for embedded hypersurfaces

Geodesic normal flow: sketch of proof

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First we could show that $\text{Vol}(\Sigma \setminus \Sigma_+) = 0$ otherwise the equality case is never achieved. This implies, by continuity, $\Sigma = \Sigma_+$ is mean convex.

Then from AM-GM we see that Σ is umbilical, and hence spherical from Nabelpunktsatz.



Willmore inequality for embedded hypersurfaces

Outreach

- This idea is recently used by Wang ¹, who gave a short proof of the Willmore inequality in Ricci nonnegative manifold established by Agostiniani-Fogagnolo-Mazzieri ².

¹Xiaodong Wang, *Remark on an inequality for closed hypersurfaces in complete manifolds with nonnegative Ricci curvature*, *Ann. Fac. Sci. Toulouse Math.*(6) (2023)

²Virginia Agostiniani, Mattia Fogagnolo and Lorenzo Mazzieri, *Sharp geometric inequalities for closed hypersurfaces in manifolds with nonnegative Ricci curvature*, *Invent. Math.*222 (2020)

Heintze-Karcher inequality

- Using geodesic normal flow which flows **into** the domain

$$\zeta(x, t) = x - t\nu(x),$$

Heintze-Karcher established the well-known Heintze-Karcher inequality for mean convex hypersurface:

$$\int_{\Sigma} \frac{1}{H} d\mu_g \geq \frac{n+1}{n} |\Omega|.$$

¹Xiaohan Jia, Guofang Wang, Chao Xia, and Xuwen Zhang, *Heintze-Karcher inequality and capillary hypersurfaces in a wedge*, arXiv:2209.13839

²_____, *Alexandrov's theorem for anisotropic capillary hypersurfaces in the half-space*, Arch. Ration. Mech. Anal. 247 (2023)

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- We ^{1 2} obtain Heintze-Karcher-type inequality for *capillary hypersurfaces* in the half-space, these hypersurfaces are critical points of the Gauss free energy functional, which meets the boundary of the container with a constant contact angle $\theta \in (0, \pi)$.

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Willmore inequality for embedded hypersurfaces in arbitrary convex domain

Asymptotic volume ratio of unbounded convex domains

- (settings) Let $K \subset \mathbf{R}^{n+1}$ be an unbounded closed convex set. Denote by $\text{Reg}(\partial K)$ the regular part of ∂K , i.e., the set of points near which ∂K can be locally written as a C^1 -hypersurface. For $x \in \text{Reg}(\partial K)$, denote by $\bar{N}(x)$ the outward unit normal to ∂K at x .

Willmore inequality for embedded hypersurfaces in arbitrary convex domain

Asymptotic volume ratio of unbounded convex domains

- (settings) Let $K \subset \mathbf{R}^{n+1}$ be an unbounded closed convex set. Denote by $\text{Reg}(\partial K)$ the regular part of ∂K , i.e., the set of points near which ∂K can be locally written as a C^1 -hypersurface. For $x \in \text{Reg}(\partial K)$, denote by $\bar{N}(x)$ the outward unit normal to ∂K at x .
- Define

$$\text{AVR}(K) = \lim_{R \rightarrow \infty} \frac{|B_R(0) \cap K|}{|B_1^{n+1}| R^{n+1}} \in [0, 1],$$

which is well-defined because the quantity $\frac{|B_R(0) \cap K|}{R^{n+1}}$ is non-increasing on $R \in [0, \infty)$, and is constant if and only if K is a convex cone with vertex at the origin.

Theorem (Jia-Wang-Xia-Z' 2024)

Let $K \subset \mathbf{R}^{n+1}$ be an unbounded closed convex set. Let $\Sigma \subset K$ be a compact, embedded C^2 -hypersurface with boundary $\partial\Sigma \subset \text{Reg}(\partial K)$ intersecting ∂K transversally such that

$$\langle \nu(x), \bar{N}(x) \rangle \geq 0, \quad \text{for any } x \in \partial\Sigma.$$

Then there holds

$$\frac{1}{n+1} \int_{\Sigma} |H|^n d\mu_g \geq \text{AVR}(K) |B_1^{n+1}|. \quad (2.1)$$

Moreover, equality in (2.1) holds if and only if Σ is a part of a sphere and $K \setminus \Omega$ is a part of the solid cone determined by Σ .

This can be viewed as a special case of its anisotropic counterpart:

Theorem (Jia-Wang-Xia-Z' 2024)

Let $K \subset \mathbf{R}^{n+1}$ be an unbounded closed convex set. Let $\Sigma \subset K$ be a compact, embedded C^2 -hypersurface with boundary $\partial\Sigma \subset \text{Reg}(\partial K)$ intersecting ∂K transversally such that

$$\langle \nu_F(x), \bar{N}(x) \rangle \geq 0, \quad \text{for any } x \in \partial\Sigma.$$

Then there holds

$$\frac{1}{n+1} \int_{\Sigma} F(\nu) |H^F|^n d\mu_g \geq \text{AVR}_F(K) |\mathcal{W}^F|. \quad (2.2)$$

Moreover, equality in (2.2) holds if and only if Σ is a part of a Wulff shape and $K \setminus \Omega$ is a part of the solid cone determined by Σ .

Main results

Application to capillary problem

- Of particular interest is the capillary phenomenon in the half-space $\mathbf{R}_+^{n+1} = \{x_{n+1} > 0\}$, which can be dated back to the study of the so-called *Sessile liquid drop problem*.

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- When trying to understand how capillarity infects boundary regularity (of local minimizers of the free energy functional), De Philippis-Maggi¹ found that in the half-space, capillarity is strongly related to anisotropy. More precisely, capillary problems can be phrased as free boundary problems in the anisotropic frameworks.

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- This fact is further revealed in the recent work by Wang-Xia².

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Main results

Application to capillary problem

- In the half-space $\mathbf{R}_+^{n+1}$, concerning capillary problem (given $\theta \in (0, \pi)$), we introduce the *capillary gauge*

$$F_\theta(\xi) = |\xi| - \cos \theta \langle \xi, E_{n+1} \rangle .$$

Clearly

$$\nabla F_\theta(\xi) = \frac{\xi}{|\xi|} - \cos \theta E_{n+1}, \nabla^2 F_\theta(\xi) = \nabla^2(|\xi|).$$

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- (some facts I)

$$\mathcal{W}_{F_\theta} = \nabla F_\theta(\mathbf{S}^n) = \mathbf{S}^n - \cos \theta E_{n+1}$$

is exactly the θ -cap (when restricted to $\mathbf{R}_+^{n+1}$).

Main results

Application to capillary problem

- (some facts II) For any set of finite perimeter $\Omega \subset \mathbf{R}_+^{n+1}$, the relative anisotropic perimeter with respect to F_θ is given by

$$P_{F_\theta}(\Omega; \mathbf{R}_+^{n+1}) = P(\Omega; \mathbf{R}_+^{n+1}) - \cos \theta P(\Omega; \partial \mathbf{R}_+^{n+1}) = \mathcal{F}_\theta(\Omega; \mathbf{R}_+^{n+1}).$$

- $\Sigma \subset \overline{\mathbf{R}_+^{n+1}}$ be a C^2 -hypersurface. Then at any $x \in \Sigma$,
 - 1 $\nu_{F_\theta}(x) = \nabla F_\theta(\nu(x)) = \nu(x) - \cos \theta E_{n+1}$, so that if Σ meets $\partial \mathbf{R}_+^{n+1}$ transversally then at any $x \in \partial \Sigma$ (capillary vs. anisotropic free boundary)

$$\langle \nu_{F_\theta}(x), E_{n+1} \rangle = \langle \nu(x), E_{n+1} \rangle - \cos \theta.$$

- 2 $d\nu_{F_\theta}|_x = d\nu|_x$, that is, the anisotropic Weingarten map is in fact the classical Weingarten map. Consequently, we have: $\kappa_i^{F_\theta}(x) = \kappa_i(x)$, and of course $H_{\Sigma, F_\theta}(x) = H_\Sigma(x)$.

Main results

Application to capillary problem

Let $K = \overline{\mathbf{R}_+^{n+1}}$ and $F = F_\theta$, we obtain:

Theorem (Jia-Wang-Xia-Z' 2024)

Given $\theta_0 \in (0, \pi)$. Let $\Sigma \subset \overline{\mathbf{R}_+^{n+1}}$ be a compact, embedded C^2 -hypersurface with boundary $\partial\Sigma \subset \partial\mathbf{R}_+^{n+1}$ intersecting $\partial\mathbf{R}_+^{n+1}$ transversally such that

$$\langle \nu, -E_{n+1} \rangle \geq -\cos \theta_0, \quad \text{for any } x \in \partial\Sigma.$$

Then there holds

$$\frac{1}{n+1} \int_{\Sigma} (1 - \cos \theta_0 \langle \nu, E_{n+1} \rangle) |H|^n d\mu_g \geq |B_{1,\theta_0}|, \quad (2.3)$$

where $B_{1,\theta_0} = B_1(-\cos \theta_0 E_{n+1}) \cap \mathbf{R}_+^{n+1}$. Moreover, equality in (2.3) holds if and only if Σ is a θ_0 -capillary spherical cap in $\overline{\mathbf{R}_+^{n+1}}$.

- 1 Willmore inequalities: an overview of the history
- 2 Willmore inequality for embedded hypersurfaces in arbitrary convex domain
- 3 Willmore inequality for capillary immersions of surfaces

Capillary immersions of surfaces in the half-space

- (immersion) Given a compact orientable smooth surface Σ , with non-empty boundary $\partial\Sigma$. Let $F : \Sigma \rightarrow \overline{\mathbf{R}_+^3}$ be an orientation preserving proper immersion.

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- (orientation and vector fields) Fix a global unit normal field ν to Σ along F , which determines an orientation on Σ and an induced orientation on $\partial\Sigma$ given by a tangential vector field τ along $\partial\Sigma$. Denote by μ the unit conormal to $\partial\Sigma$ in Σ so that $\{\tau, \nu, \mu\}$ is compatible with $\{E_1, E_2, E_3\}$ of $\mathbf{R}^3$. Let $\bar{\nu}$ be the unit normal to $\partial\Sigma$ in $\partial\mathbf{R}_+^3$ so that $\{\bar{\nu}, -E_3\}$ compatible with $\{\nu, \mu\}$.

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- (capillary immersion) Given $\theta \in (0, \pi)$, we say that the proper immersion $F : \Sigma \rightarrow \overline{\mathbf{R}_+^3}$ is a *capillary immersion* (or θ -capillary immersion) with contact angle θ if the angle determined by μ and $\bar{\nu}$ is constant and equals to θ along $\partial\Sigma$, that is, for any $x \in \partial\Sigma$,

$$\mu(x) = \sin \theta(-E_3) + \cos \theta \bar{\nu}(x).$$

Capillary immersions of surfaces in the half-space

- (signed area) For the immersion $F : \partial\Sigma \rightarrow \partial\mathbf{R}_+^3$, we use T to denote the “enclosed domain” by $F(\partial\Sigma)$, which means the topological boundary ∂T is given by $F(\partial\Sigma)$ and it has induced orientation by τ . We use $|T|$ to denote the oriented area of T , which is defined to be

$$|T| = \sum_i \operatorname{sgn}(T_i) |T_i|,$$

where each T_i is a bounded domain such that ∂T_i is a simply closed curve and $\operatorname{sgn}(T_i) = +1$ if $\bar{\nu}$ points outward T_i , while $\operatorname{sgn}(T_i) = -1$ if $\bar{\nu}$ points inward T_i . $|T|$ is the signed area of the so-called “wetting part”.

Willmore energy of capillary immersions in the half-space

- The Willmore energy $\mathcal{W}(F)$ of a smooth immersed compact connected orientable surface $F : \Sigma \rightarrow \mathbf{R}^3$ with boundary $\partial\Sigma$, which may have several connected components, is usually proposed by

$$\mathcal{W}_0(F) := \frac{1}{4} \int_{\Sigma} |\mathbf{H}_{\Sigma}|^2 d\mu_g + \int_{\partial\Sigma} \kappa_g.$$

Thanks to the well-known Gauss-Bonnet theorem, the Willmore energy may be rewritten as

$$\mathcal{W}_0(F) = \frac{1}{2} \int_{\Sigma} |\mathring{A}|^2 d\mu_g + 2\pi\chi(\Sigma),$$

which is conformal invariant.

Willmore energy of capillary immersions in the half-space

- For any θ -capillary immersion $F : \Sigma \rightarrow \overline{\mathbf{R}_+^3}$, it holds locally

$$\kappa_g = \nabla_\tau \mu_\Sigma \cdot \tau = \nabla_\tau (\cos \theta \bar{\nu} - \sin \theta E_3) \cdot \tau = \cos \theta \tilde{\kappa}_g,$$

where $\tilde{\kappa}_g$ is the geodesic curvature of $F(\partial\Sigma)$ as an immersion in $\partial\mathbf{R}_+^3$. Again, by Gauss-Bonnet,

$$\int_{\partial\Sigma} \kappa_g = \cos \theta \int_{\partial\Sigma} \tilde{\kappa}_g = 2\pi \cos \theta \operatorname{ind}(F(\partial\Sigma)),$$

where $\operatorname{ind}(F(\partial\Sigma))$ is rotation index of immersed plane curve $F(\partial\Sigma)$, which is again a topological invariant.

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- This shows that

$$\frac{1}{4} \int_\Sigma |\mathbf{H}_\Sigma|^2 d\mu_g = \mathcal{W}_0(F) - \int_{\partial\Sigma} \kappa_g = \mathcal{W}_0(F) - 2\pi \cos \theta \operatorname{ind}(F(\partial\Sigma))$$

is conformal invariant, and we define it to be the Willmore energy of the capillary immersion F , denoted by $\mathcal{W}(F)$.

Main result: Li-Yau-type inequality

Theorem (Li-Yau-type inequality (I). Wang-Xia-Z' 2024)

Given $\theta \in (0, \pi)$, let $F : \Sigma \rightarrow \overline{\mathbf{R}^3_+}$ be a compact θ -capillary immersion, satisfying

$$\text{wind}(x) \geq 0 \text{ for any } x \in \partial\mathbf{R}^3_+,$$

where $\text{wind}(x)$ is the winding number of $F(\partial\Sigma)$ about $x \in \partial\mathbf{R}^3$. Then we have

① the Li-Yau-type inequality

$$\mathcal{W}(F) \geq 4(1 - \cos \theta)\pi\Theta^2(\mu, x_0)$$

holds for every $x_0 \in F(\partial\Sigma)$, where $\Theta^2(\mu, x)$ denotes the density of μ at x_0 .

Main result: Li-Yau-type inequality

Theorem (Li-Yau-type inequality (II). Wang-Xia-Z' 2024)

- ① *the sharp estimate on Willmore energy:*

$$\mathcal{W}(F) \geq 2(1 - \cos \theta)\pi, \quad (3.1)$$

and if

$$\mathcal{W}(F) < 4(1 - \cos \theta)\pi,$$

then $F : \partial\Sigma \subset \partial\mathbf{R}_+^3$ must be an embedding.

- ② *For $\theta \in [\frac{\pi}{2}, \pi)$, if*

$$W(F) < 4\pi,$$

then $F : \Sigma \rightarrow \overline{\mathbf{R}_+^3}$ is an embedding.

Moreover, equality in (3.1) holds if and only if $F(\Sigma)$ is a θ -spherical cap.

Main result: Li-Yau-type inequality

- The Li-Yau-type inequality follows from the establishment of a Simon-type Monotonicity identity.

Main result: Li-Yau-type inequality

- The Li-Yau-type inequality follows from the establishment of a Simon-type Monotonicity identity.
- The condition $w(x) \geq 0$ is essential in proving the monotonicity, which can be seen from the following example:

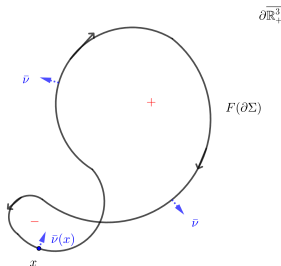


Figure: Example.

- Due to the conformal invariance of the Willmore functional, for any capillary immersions of surfaces into $\mathbb{B}^3$, the Willmore energy is still

$$\mathcal{W}(F) := \frac{1}{4} \int_{\Sigma} |\vec{\mathbf{H}}|^2 d\mu_g + \int_{\partial\Sigma} \kappa_g - 2\pi \cos \theta \operatorname{ind}(F(\partial\Sigma)).$$

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- The geodesic curvature is given by

$$\kappa_g = \nabla_{\tau} \mu_{\Sigma} \cdot \tau = \nabla_{\tau} (\cos \theta \bar{\nu} + \sin \theta \bar{N}(x)) \cdot \tau = \cos \theta \tilde{\kappa}_g + \sin \theta,$$

where $\bar{N}(x) = x$ is the outer unit normal of $\mathbb{B}^3$ and $\tilde{\kappa}_g$ is the geodesic curvature of $F : \partial\Sigma \rightarrow \mathbf{S}^2$. By Gauß-Bonnet we have

$$2\pi \operatorname{ind}(F(\partial\Sigma)) = \int_{\partial\Sigma} \tilde{\kappa}_g + |T|,$$

where $|T|$ is the oriented area of T . Altogether implying that

$$\mathcal{W}(F) = \frac{1}{4} \int_{\Sigma} |\vec{\mathbf{H}}|^2 d\mu_g + \sin \theta |\partial\Sigma| - \cos \theta |T|.$$

Willmore inequality for capillary immersions in $\mathbb{B}^3$

Corollary (Wang-Xia-Z' 2024)

Given $\theta \in (0, \pi)$, let $F : \Sigma \rightarrow \overline{\mathbb{B}^3}$ be a θ -capillary immersion with non-negative winding number. Then, there holds

$$\frac{1}{4} \int_{\Sigma} |\vec{\mathbf{H}}|^2 d\mu_g + \sin \theta |\partial \Sigma| - \cos \theta |T| = \mathcal{W}(F) \geq 2(1 - \cos \theta)\pi.$$

Equality holds if and only if $F(\Sigma)$ is a spherical cap or a totally geodesic disk in $\overline{\mathbb{B}^3}$.

Corollary (Wang-Xia-Z' 2024)

Under the above conditions, if the immersion is minimal, then there holds

$$2|\Sigma| - \cos \theta |T| \geq 2(1 - \cos \theta)\pi.$$

Equality holds if and only if $F(\Sigma)$ is a totally geodesic disk in $\overline{\mathbb{B}^3}$.

Willmore inequality for capillary immersions in $\mathbf{R}_+^3$ or $\mathbb{B}^3$

Some remarks

- Although the results in the Euclidean ball case can be obtained from conformal invariance, we also establish a Simon-type Monotonicity identity in $\mathbb{B}^3$, which gives alternative proofs of the listed results.

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- Before our work, the Willmore inequality for capillary surfaces in $\mathbf{R}_+^3$ was proved using a flow method by Wang-Weng-Xia¹ provided that the surfaces are convex and the contact angle θ is restricted within the range $(0, \frac{\pi}{2}]$.

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- Before our work, the Willmore inequality for capillary surfaces in $\mathbf{R}_+^3$ was proved using a flow method by Wang-Weng-Xia ¹ provided that the surfaces are convex and the contact angle θ is restricted within the range $(0, \frac{\pi}{2}]$.
- The following optimal area estimate for free boundary ($\theta = \frac{\pi}{2}$) minimal $\Sigma^k \subset \mathbb{B}^{n+1}$ is proven by Fraser-Schoen ² and Brendle ³

$$|\Sigma^2| \geq \pi, \quad |\Sigma^k| \geq |\mathbb{B}^k|.$$

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Willmore inequality for capillary immersions in $\mathbf{R}_+^3$ or $\mathbb{B}^3$

Some remarks

- As a special case of the isoperimetric inequality for minimal submanifolds proved by Brendle¹, any capillary minimal surface Σ in the unit ball satisfies the following optimal estimate

$$|\Sigma| \geq \sin^2 \theta \pi = \sin^2 \theta |\mathbb{B}^2|$$

with equality holds if and only if Σ is a flat disk. (higher dimensional counterparts also hold)

- This estimate and our estimate does not imply each other.

¹Simon Brendle, *Minimal hypersurfaces and geometric inequalities*, *Ann. Fac. Sci. Toulouse Math.* (6)32 (2023)

Thank you very much for your attention!